Unsupervised learning

Instructor: Chao Huang

Department of Computer Science
Montclair State University

Main sources: CBMM 9.54 @MIT

Supervised learning vs. Unsupervised Learning

- Supervised learning
  - Discover patterns in data that relate features with a target value
  - Patterns are used to predict the values of the target values in future data samples
- Unsupervised learning
  - Data have no target values
  - Aim to explore the data to find some intrinsic structures

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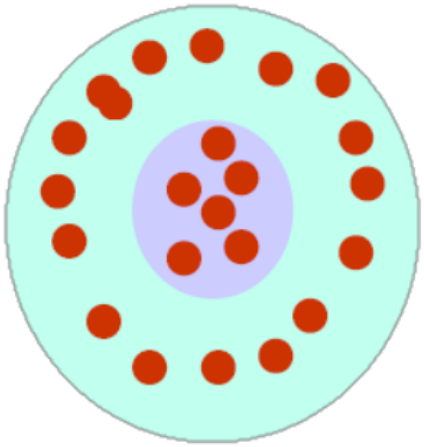
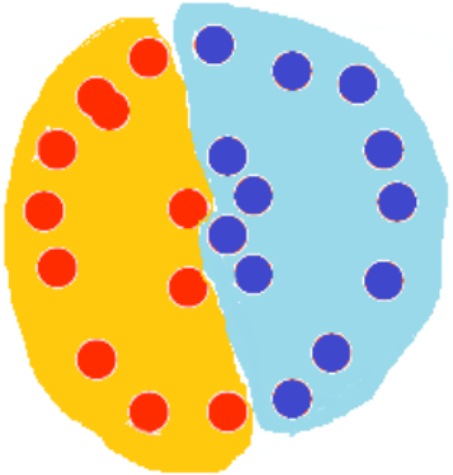
Types of Unsupervised Learning

- **Clustering**
  - Group data into sets of items similar to each other
  - Example
    - ★ Group customers based on purchasing patterns
- **Unsupervised transformation**
  - Create new data representations easier to understand
  - Example
    - ★ Dimensionality reduction of a gene dataset to a few key components

What is Clustering

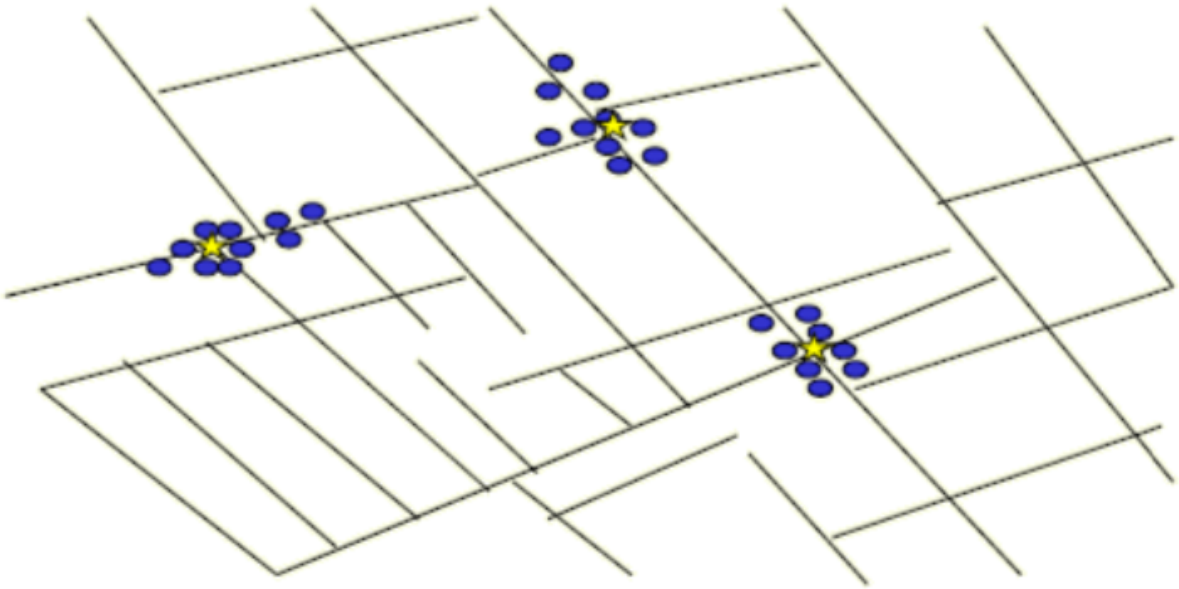
- Organization of unlabeled data into similarity groups called clusters
- A cluster is a collection of data items which are "similar" between them, and "dissimilar" to data items in other clusters





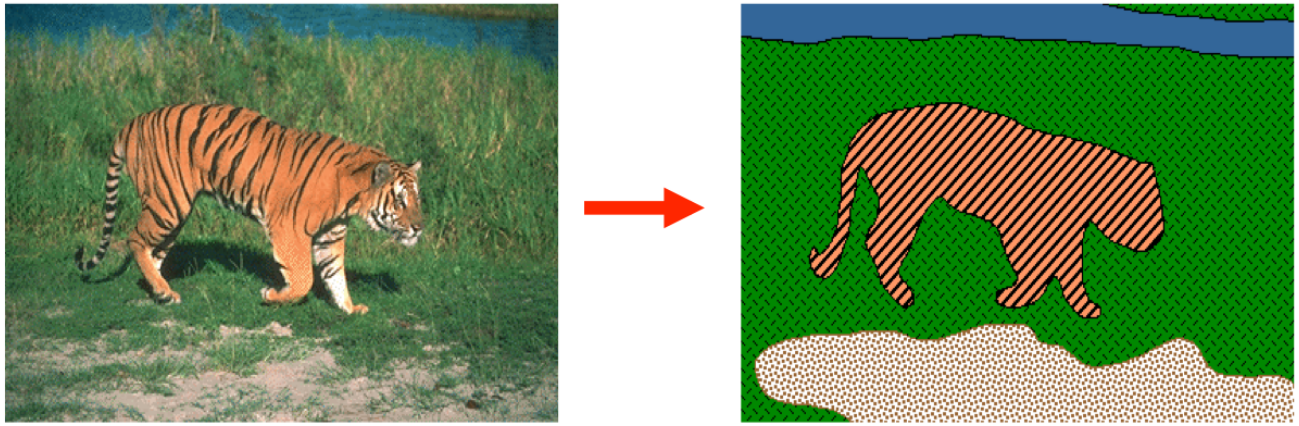
A Story of Clustering

- John Snow, a London physician, plotted the location of cholera deaths on a map during an outbreak in 1850s
- The locations revealed that cases were clustered around certain intersections where there were polluted wells (which exposes both the problem and solution)

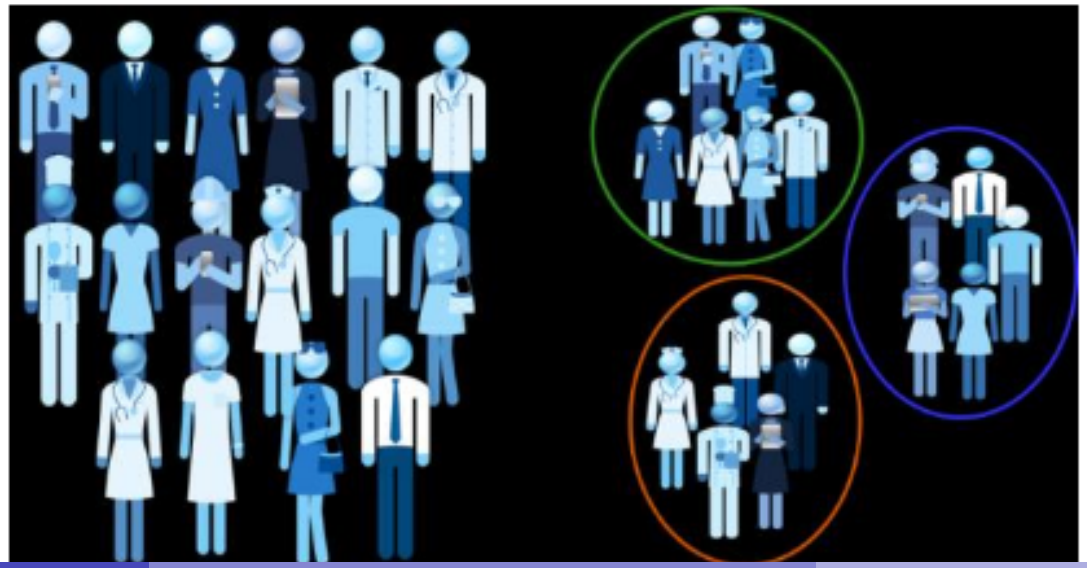


Clustering Applications

- Computer vision, e.g., segmentation



- Insurance, e.g., identifying groups of motor insurance policy holders with low/medium/high average claim cost



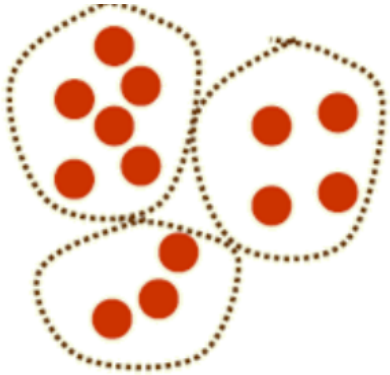
Clustering Applications - Cont'd

- City planning, e.g., identifying groups of houses according to their house type, value, and geographical location

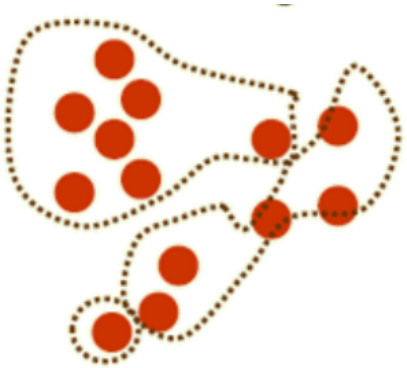
- Earthquake studies, e.g., observed earthquake epicenters should be clustered among continent faults
- Many more $\cdots\cdots$

What Do We Need for Clustering?

1. Proximity measure, either
   - Similarity measure $s(x_i, x_j)$: large if x_i, x_j are similar
   - Dissimilarity/distance measure $d(x_i, x_j)$: small if x_i, x_k are similar
2. Criterion function to evaluate a clustering



good clustering



bad clustering

3. Algorithm to compute clustering

Distance Measures

- Euclidean distance

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^{d} \left(x_i^{(k)} - x_j^{(k)} \right)^2}$$

- Manhattan (city block) distance

$$d(x_i, x_j) = \sum_{k=1}^{d} \left| x_i^{(k)} - x_j^{(k)} \right|$$

- Both are special cases of Minkowski distance:

$$d_p(x_i, x_j) = \left(\sum_{k=1}^{d} \left| |x_i^{(k)} - x_j^{(k)}| \right|^p \right)^{1/p}$$

  - p is a positive integer

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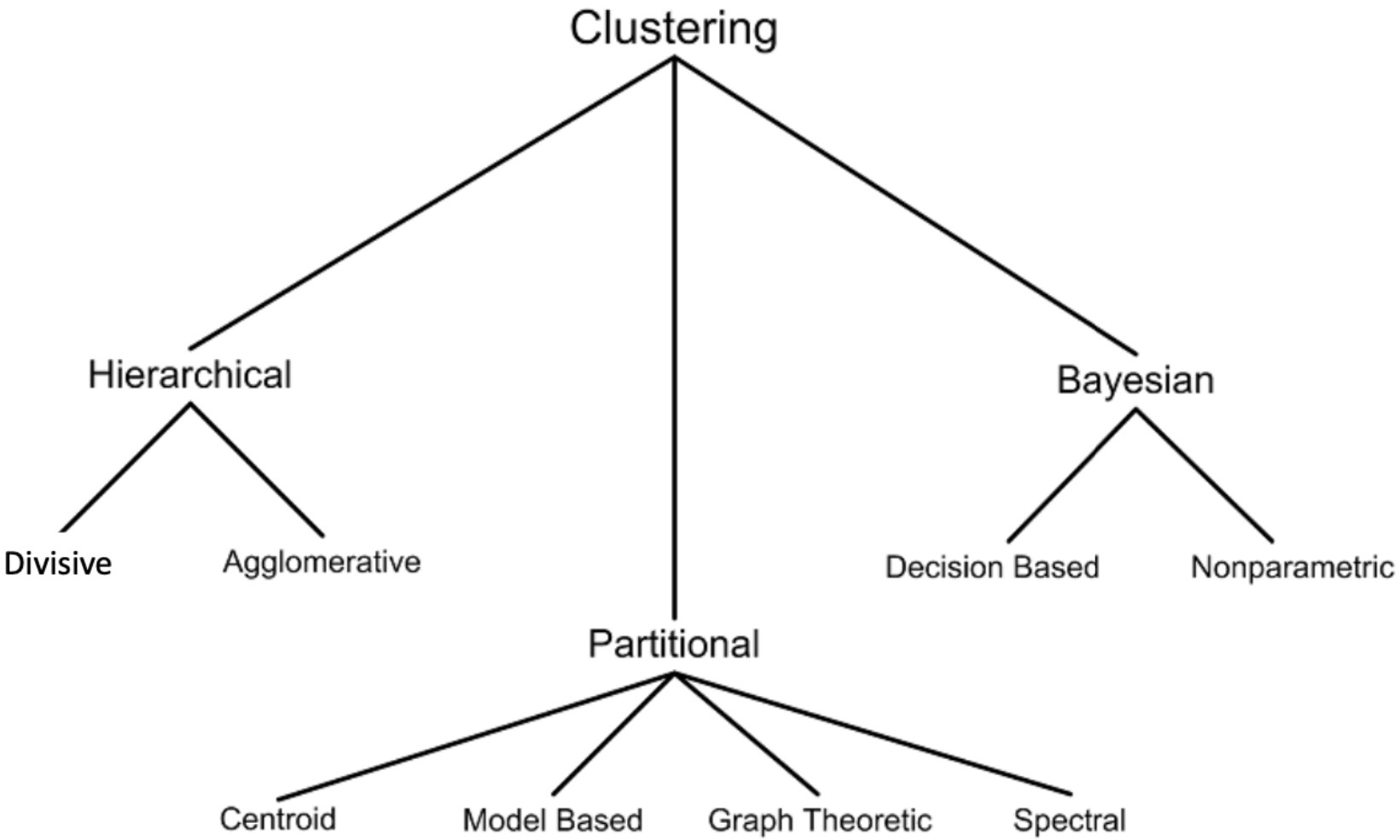
Cluster Evaluation

- Intra-cluster cohesion (compactness)
  - Cohesion measures how near the data samples in a cluster are to the cluster centroid
  - Sum of squared error (SSE) is a commonly used measure
- Inter-cluster separation (isolation)
  - Different cluster centroids should be far away from one another
- Cluster evaluation is a hard problem
  - Domain knowledge and expert judgements are still the key

Clustering Techniques



Clustering Techniques

- Hierarchical algo. find successive clusters using previous clusters
  - Agglomerative algo. begin with each point as a separate cluster and merge them into successively large clusters
  - Divisive algo. begin with the entire set and proceed to divide it into successively smaller clusters
- Partitional algo. divides data into distinct non-overlapping clusters, where each data point belongs to exactly one cluster
- Bayesian algo. try to generate a posterior distirbution over the collection of all partitions of the data

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K-Means Clustering

- Widely used, by MacQueen 1967
- Consider dataset $\mathcal{D}$: $\{\boldsymbol{x}_1, \boldsymbol{x}_2 \cdots, \boldsymbol{x}_n\}$
- K-means algo. partitions the given data into k clusters
  - k specified by user
  - Each cluster has a cluster center/centroid

K-Means Algorithm

- Given k, k-means works as follows:
  1. Choose k random data points (seeds) to be the initial centriods
  2. Assign each data point to the closest centroid
  3. Re-compute the centroids using the current cluster memberships
  4. Check convergence. If a convergence criterion is not met, repeat steps 2 and 3

K-Means Convergece Criterion

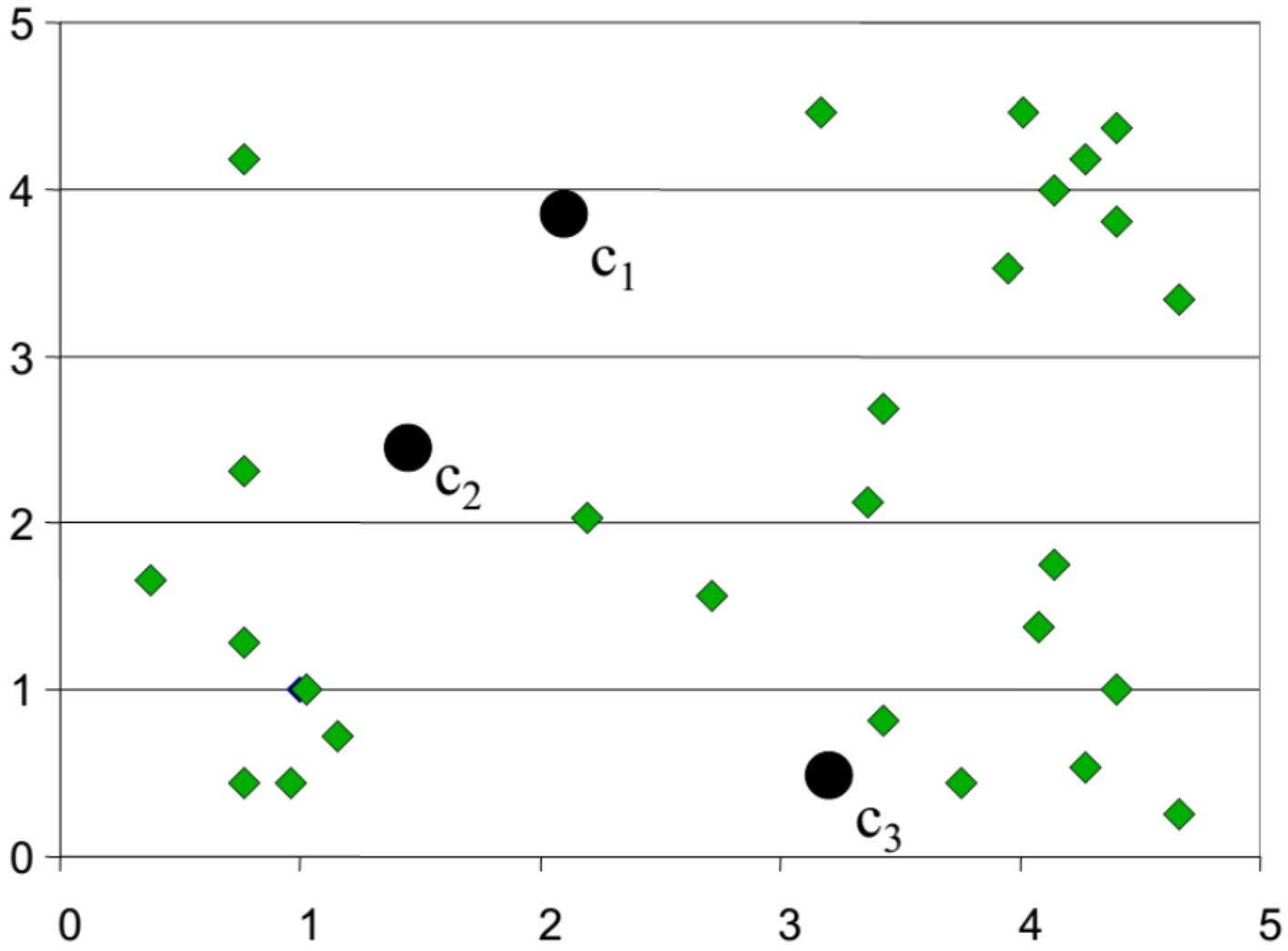
- No (or minimum) re-assignment of data points to different clusters, or
- No (or minimum) change of centroid, or
- Minimum decrease in the sum of squared error (SSE)

$$SSE = \sum_{j=1}^{k} \sum_{\boldsymbol{x} \in C_j} d(\boldsymbol{x}, \boldsymbol{m}_j)^2$$

  - C_j: the j-th cluster
  - $\boldsymbol{m}_j$: centroid of cluster C_j, i.e., mean vector of all data points in C_j
  - $d(\boldsymbol{x}, \boldsymbol{m}_j)$: (Euclidean) distance between data point $\boldsymbol{x}$ and centroid $\boldsymbol{m}_j$

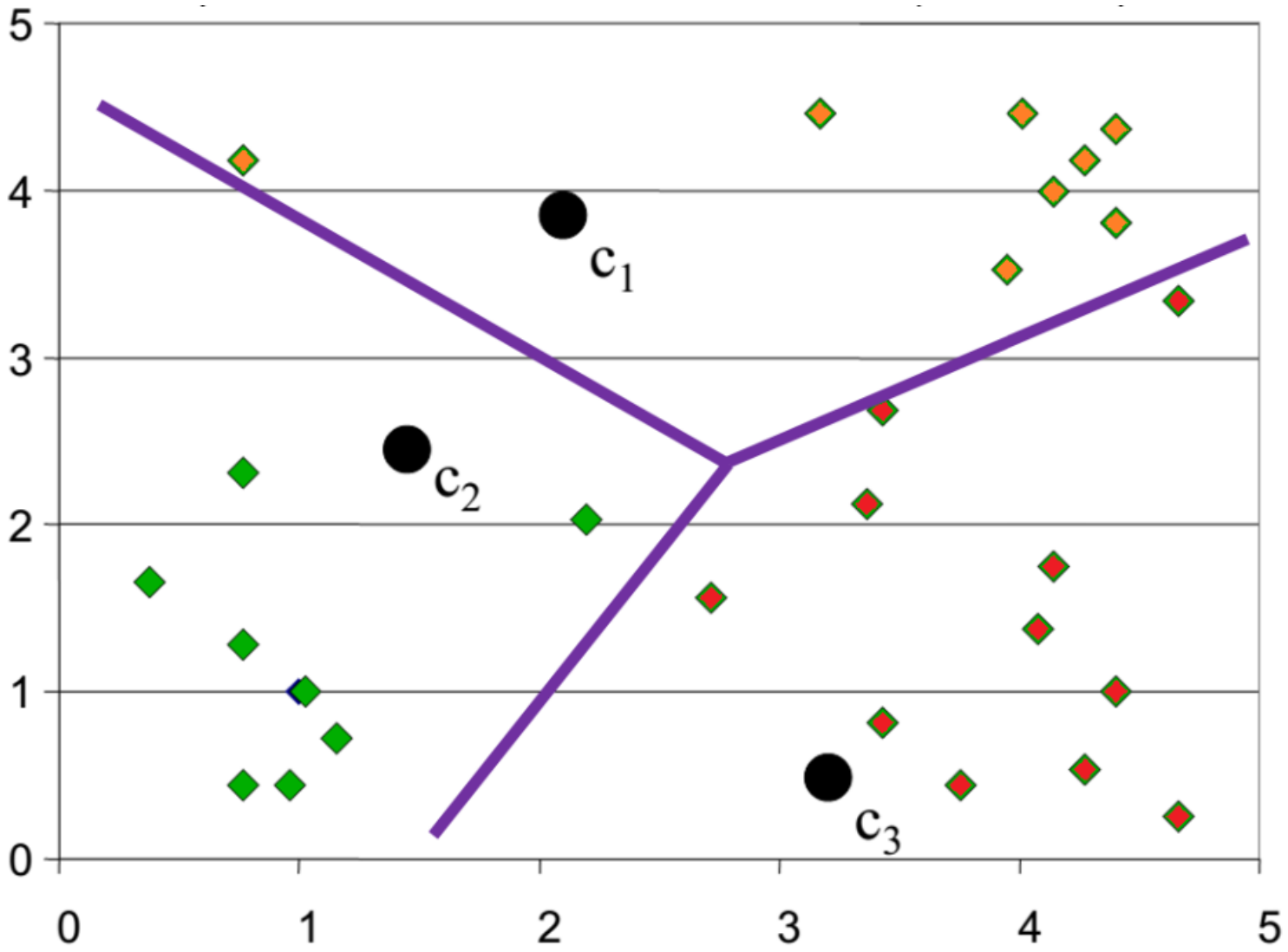
K-means Example

- Step 1: randomly choose k points to be centroids



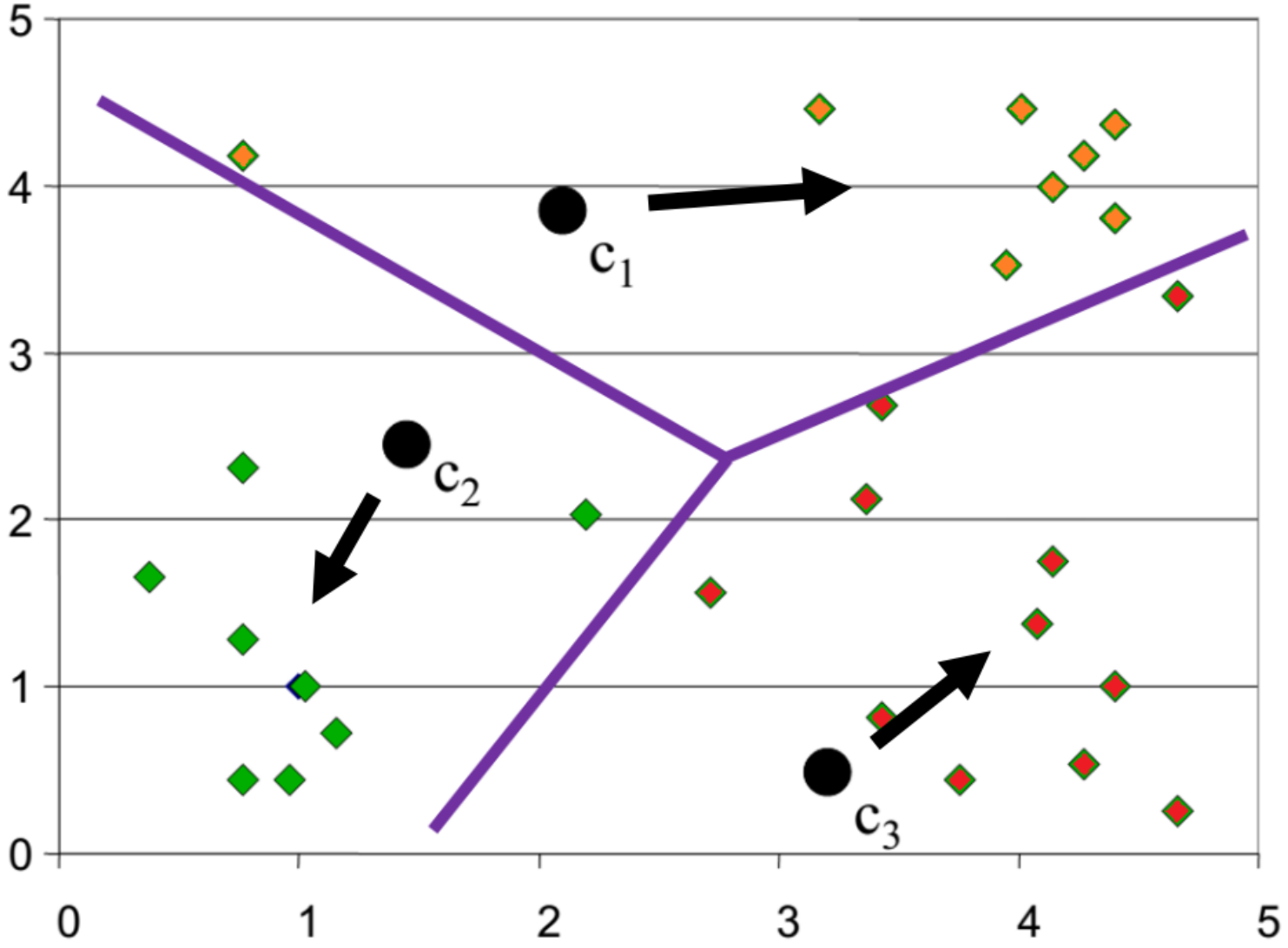
K-means Example

- Step 2: assign each data point to closest centroid



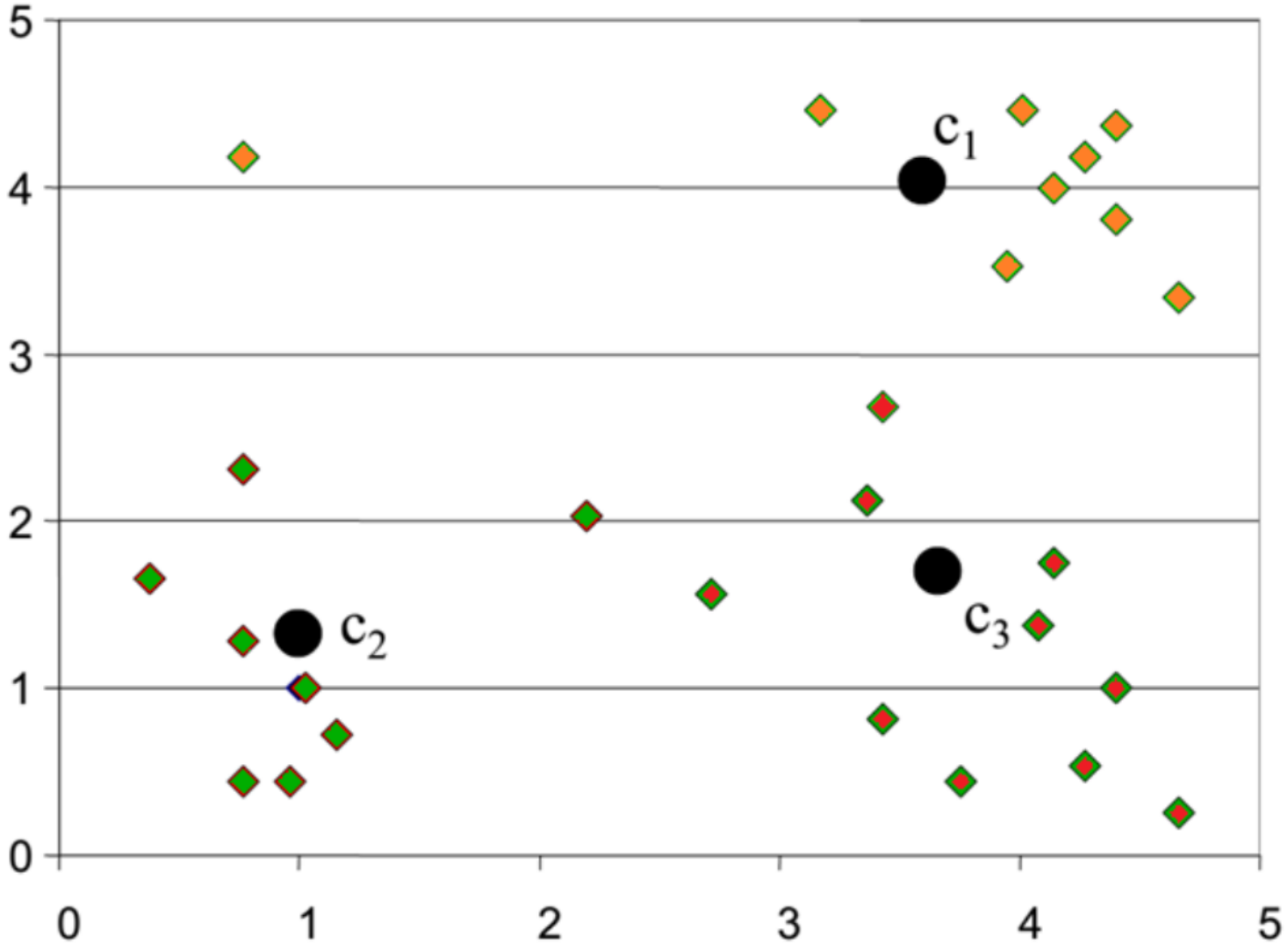
K-means Example

- Step 3: recompute the centroids using current cluster memberships



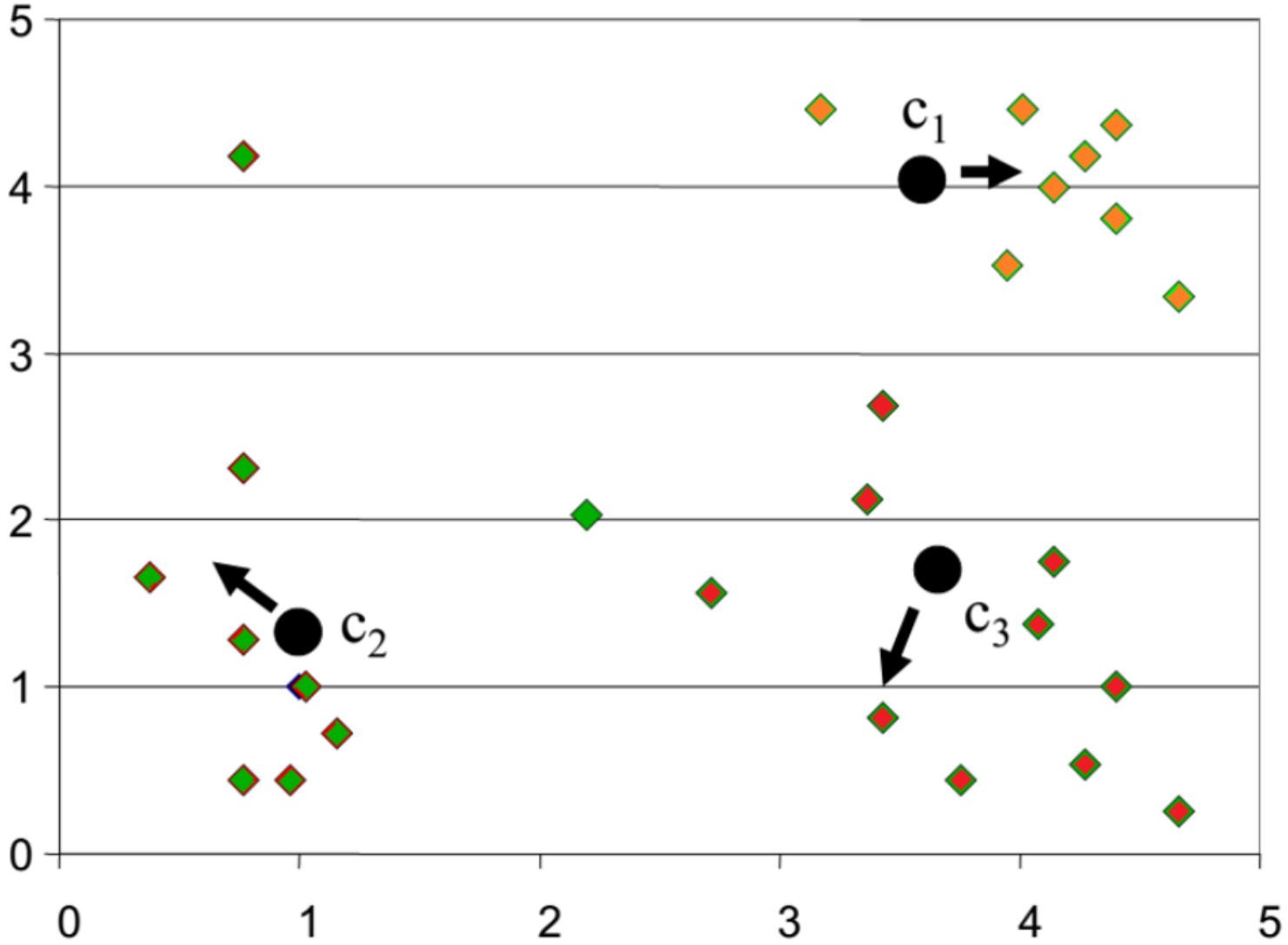
K-means Example

- Result of first iteration



K-means Example

- Second iteration



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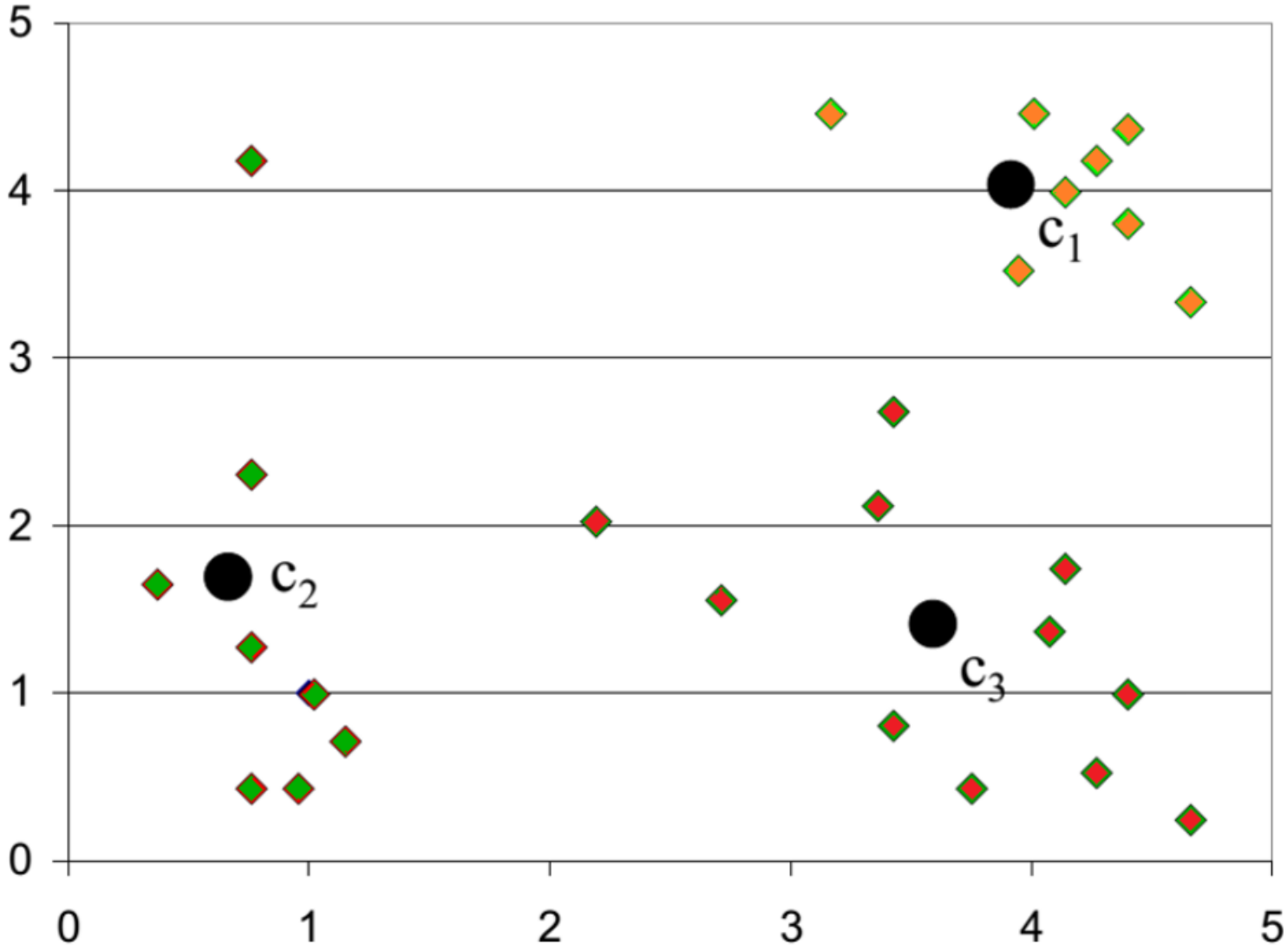

Discussion of K-Means

- Pros
  - Easy to understand and implement; most popular clustering algo.
  - Efficient: time complexity $O(tkn)$
    - n: # data points
    - k: # clusters
    - t: # iterations
  - Since both k and t are small, k-maens is considered a linear algorithm
- Cons
  - The algo. is applicable only if the **mean** is defined
    - For categorical data, k-mode: the centroid is represented by most frequent values
  - Choice of k can be tricky
  - Algo. is sensitive to **outliers**
    - Outliers are data points that are very far away from other data points
    - Outliers could be errors in the data recording or some special data points with very different values

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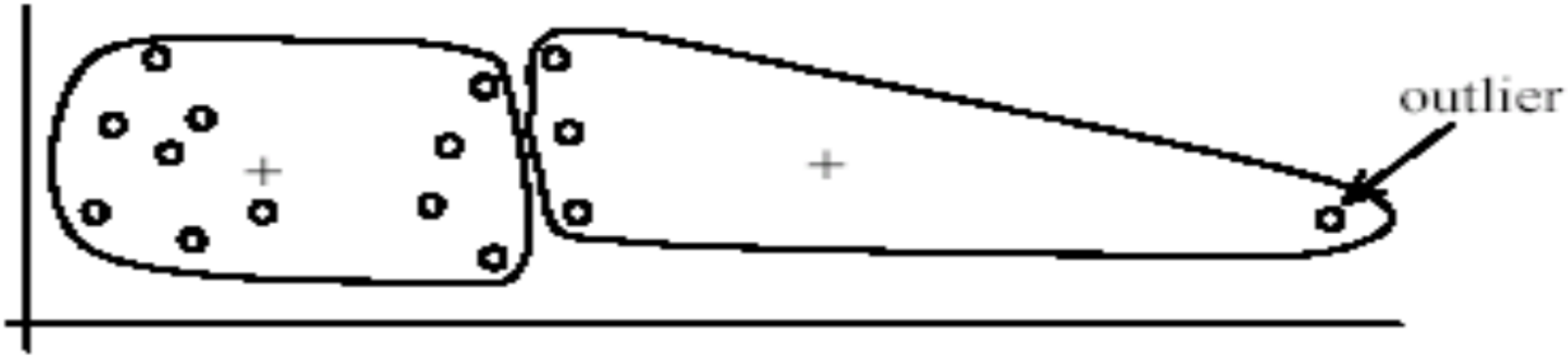
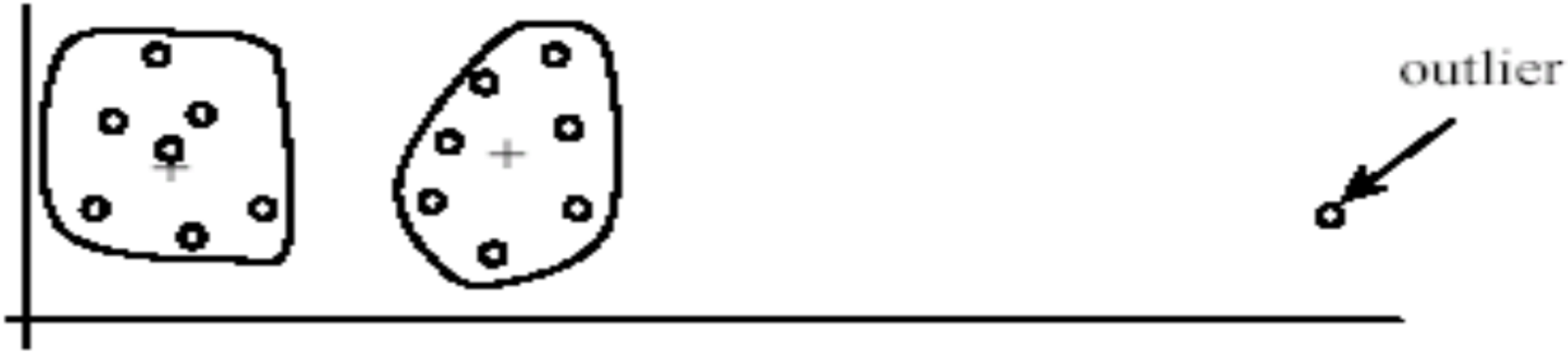
Outliers



(A): Undesirable clusters

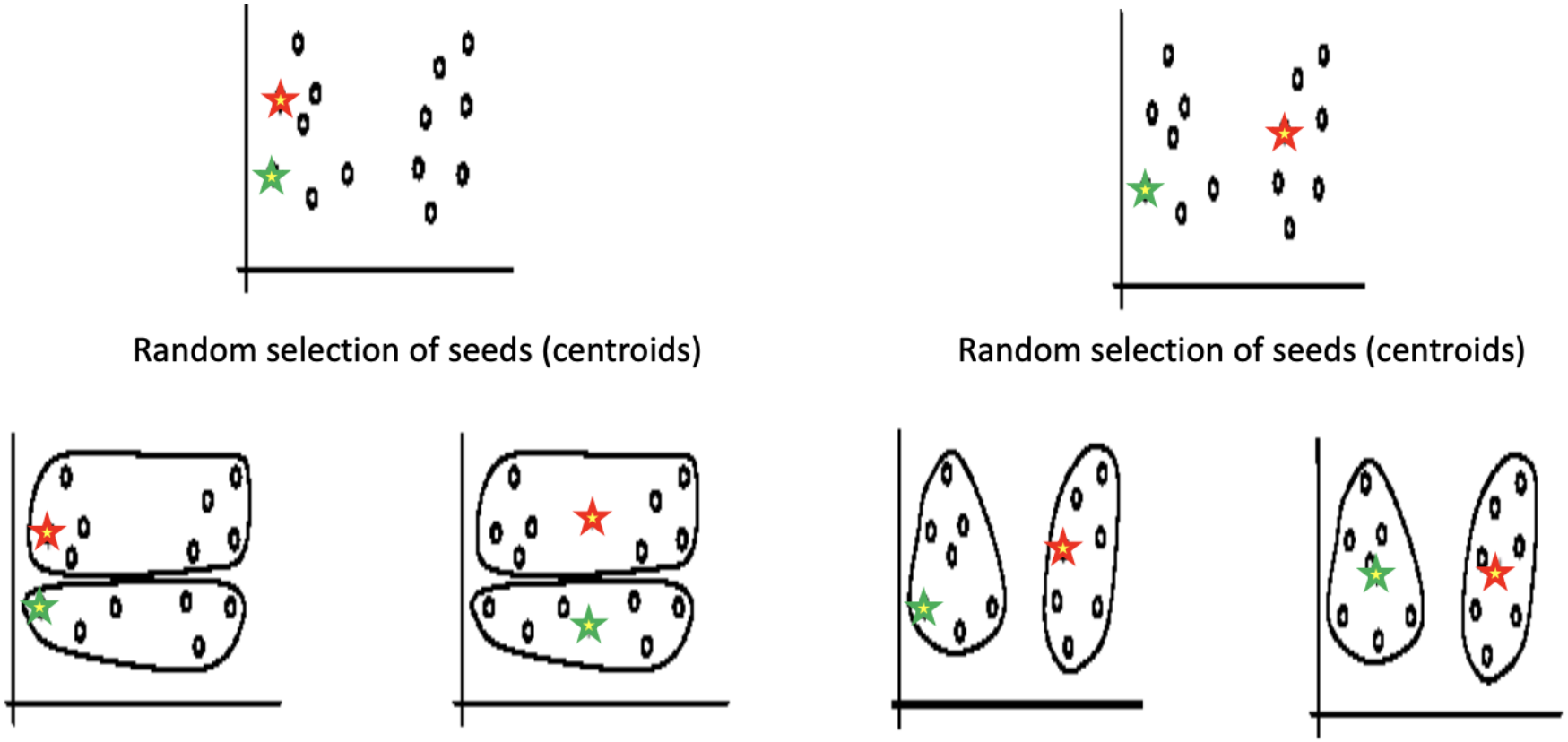


(B): Ideal clusters

Dealing with Outliers

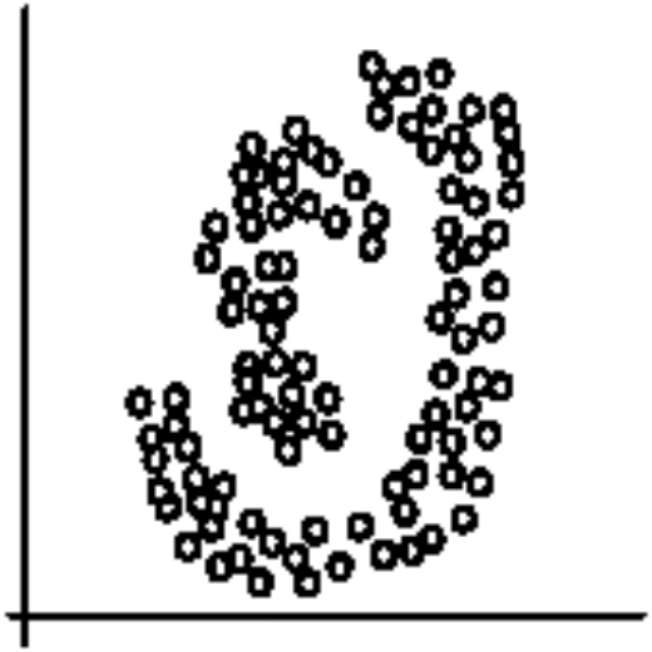
- Remove some data points that are much further away from the centroids than other data points
  - To be safe, we may want to monitor these possible outliers over a few iterations and then decide to remove them
- Sampling a (random) subset of data points, so that the chance of selecting an outlier is smaller
  - Assign the rest of the data points to the clusters by distance

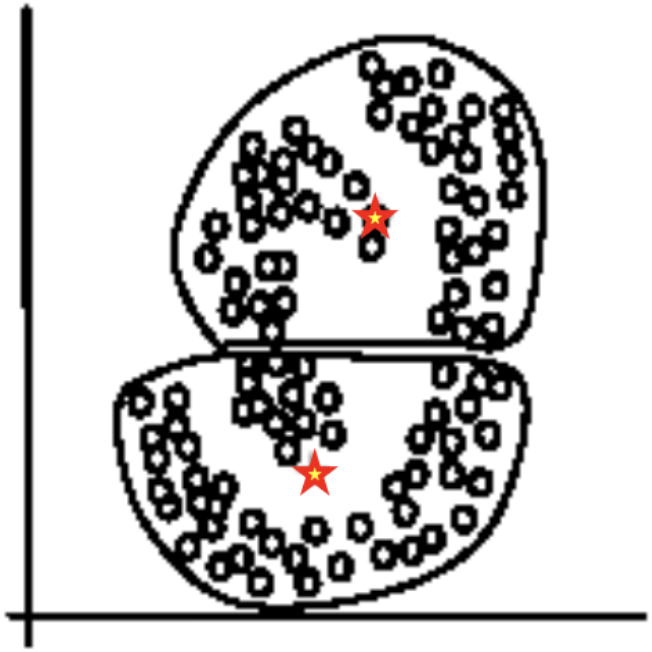
Sensitivity to Initial Centroids



Special Structures

- K-means not suitable for discovering clusters that are not hyper-ellipsoids/hyper-spheres





Discussion

- K-means still popular due to its simplicity and efficiency
- No proof that any other clustering algo. performs better in general
- Comparing different clustering algo. can be challenging. No one knows the "correct" clusters
- Question: is k-means a hierarchical, partitional, or Bayesian algo?

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Hierarchical Clustering

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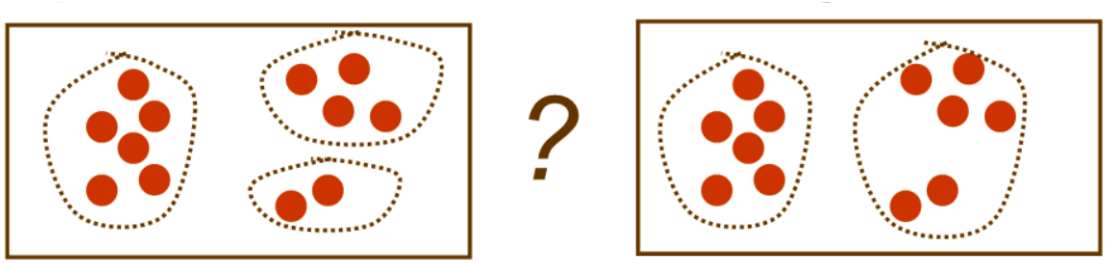
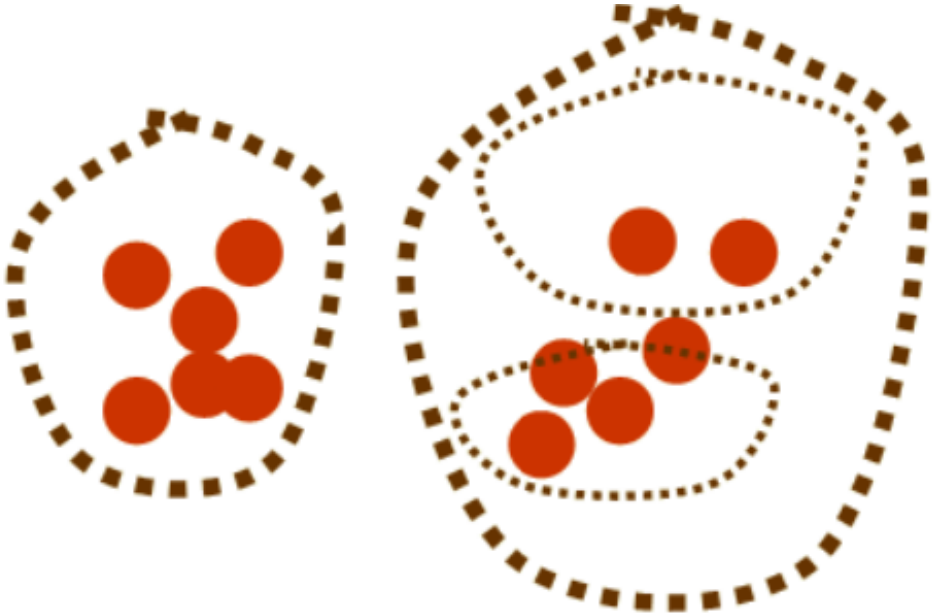


- For some data, hierarchical clustering is more appropriate than partitional clustering



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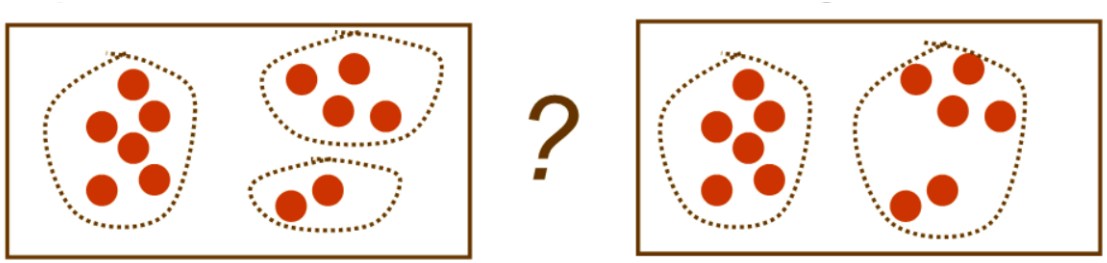

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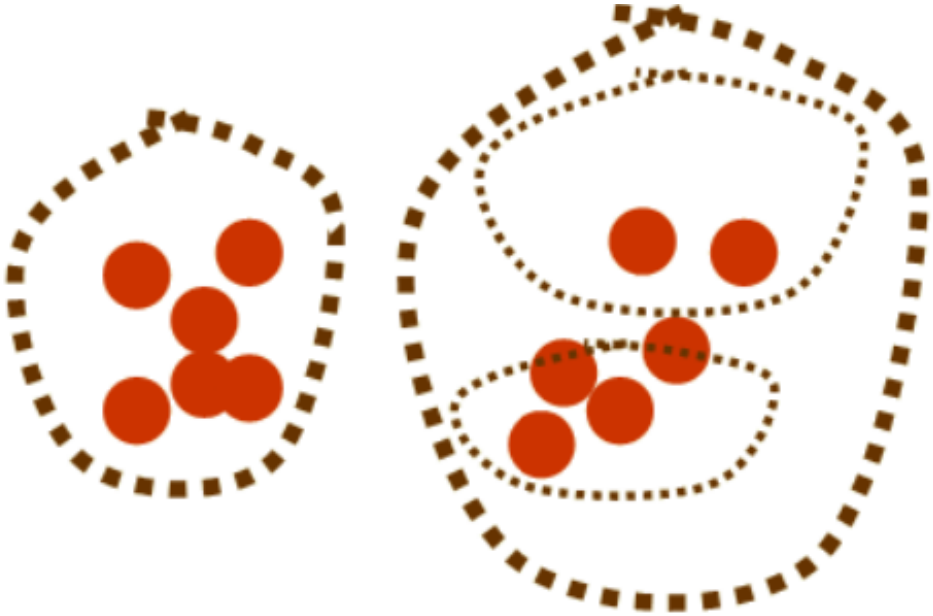
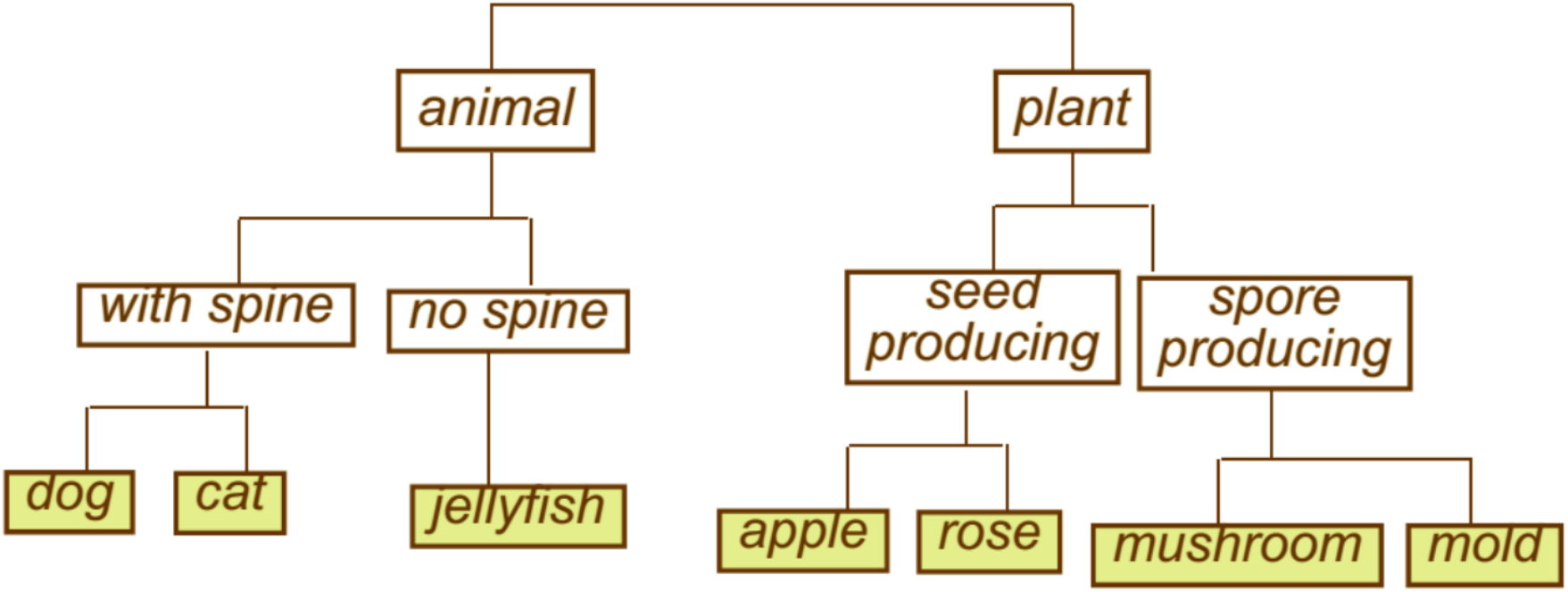
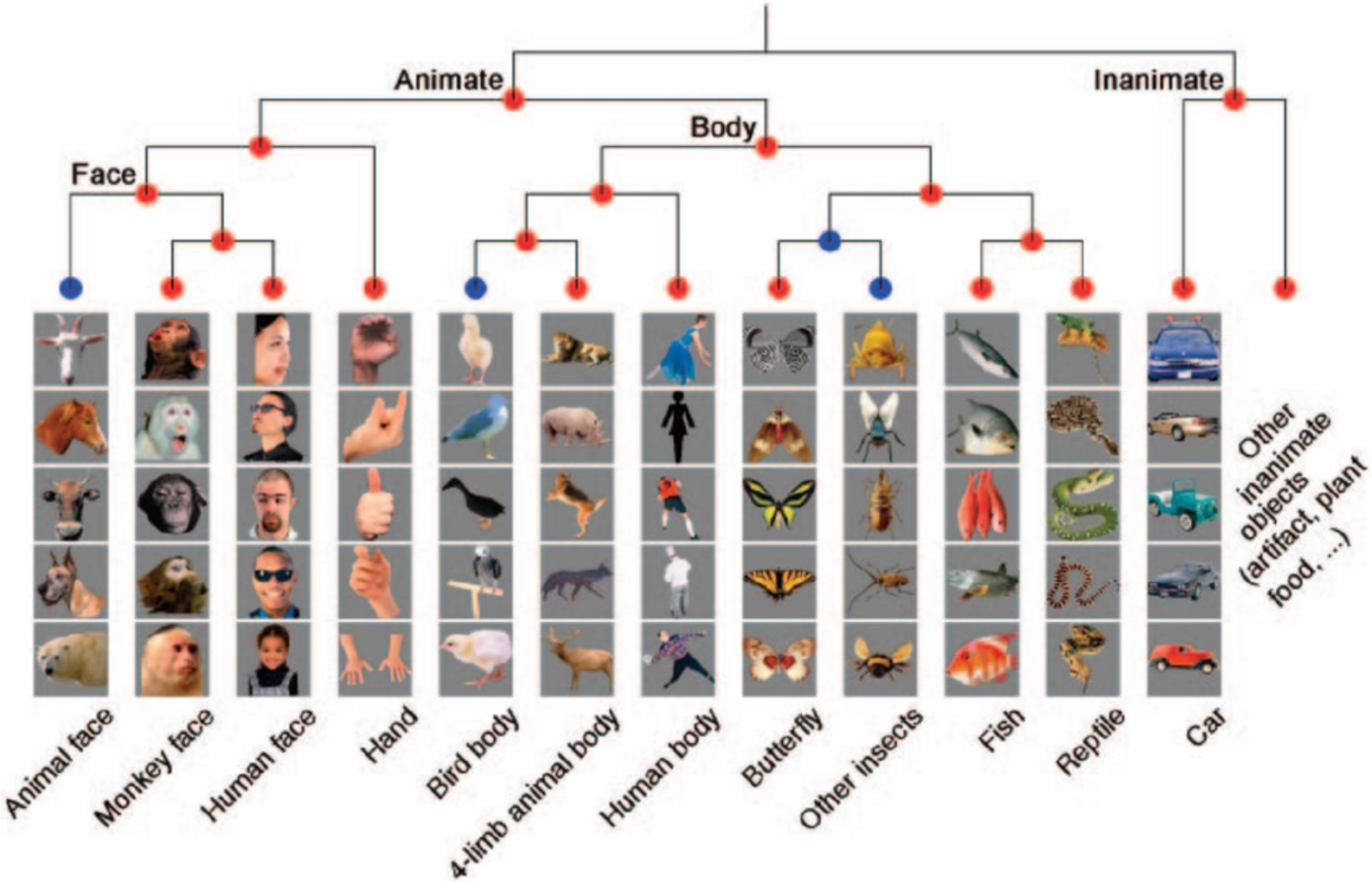


Example: Biological Taxonomy



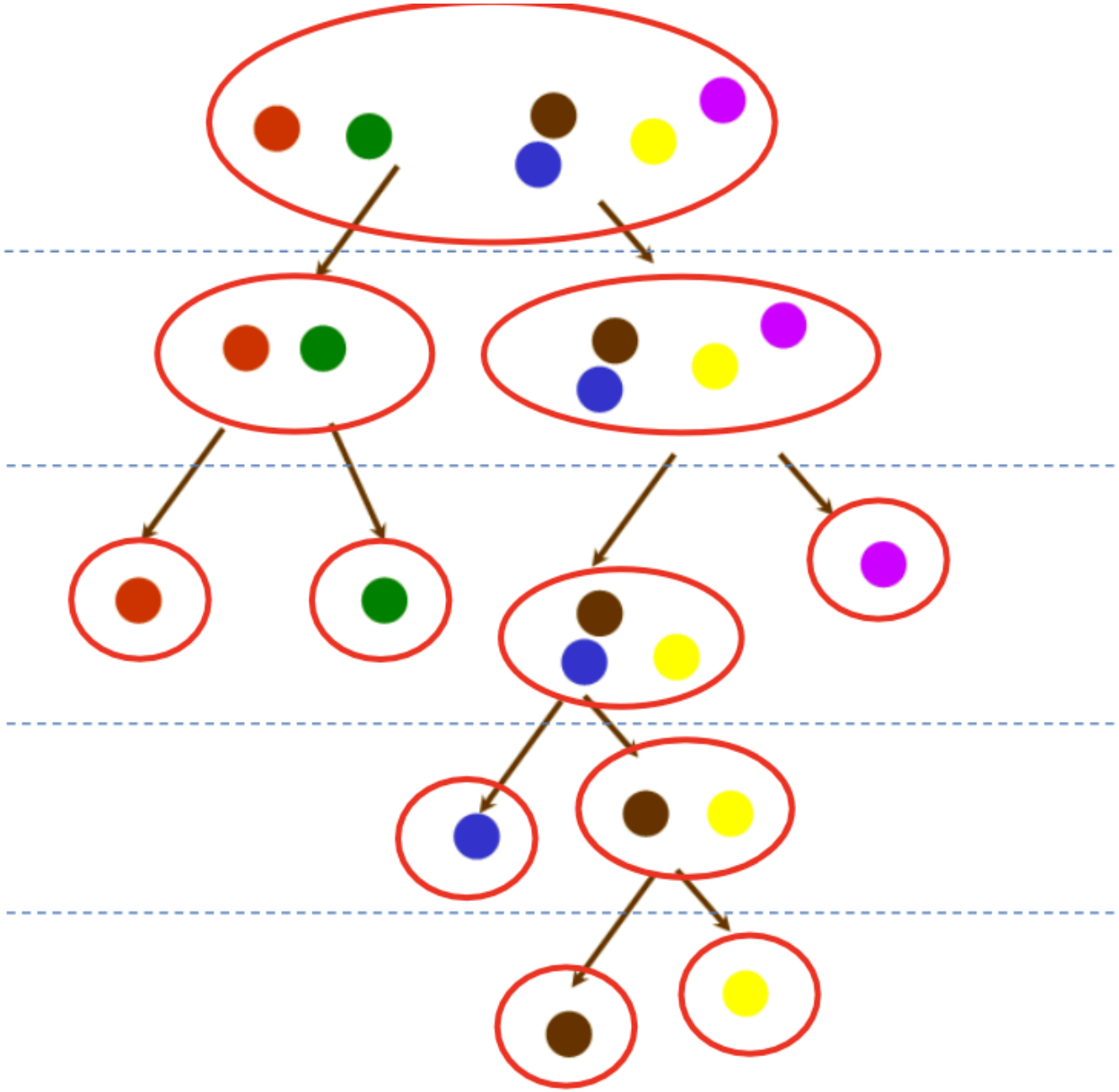
Example: Neural Response



Types of Hierarchical Clustering

- Divisive clustering (top down) starts with all data points in one cluster, the root, then
  - splits the roots into a set of child clusters. Each child cluster is recursively divided further
  - stops when only singleton clusters of individual data points remain, i.e., each cluster only has a single point
- Agglomerative clustering (bottom up). The dendrogram is built from bottom level. It
  - merges the most similar (nearest) pairs of clusters
  - stops when all data points are merged into a single cluster

Divisive Hierarchical Clustering

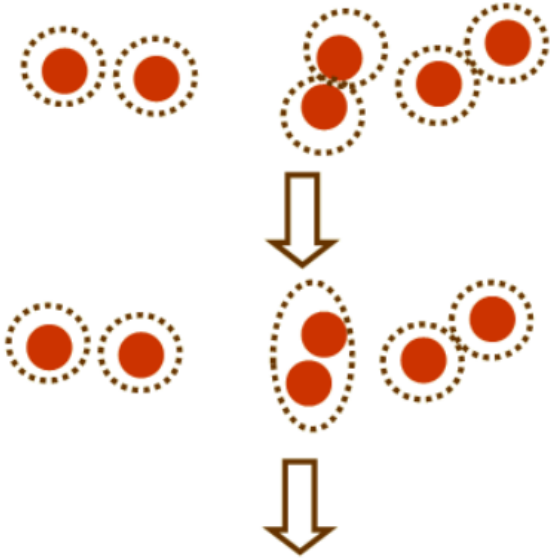


Agglomerative Hierarchical Clustering

initialize with each example in singleton cluster

while there is more than ***1*** cluster

1. find 2 nearest clusters
2. merge them



Summary

- Unsupervised learning
- K-means clustering
- Hierarchical clustering
  - Divisive
  - Agglomerative

After-Class Reading

- Müller A C, Guido S. Introduction to machine learning with Python: a guide for data scientists[M]. " O'Reilly Media, Inc.", 2016.
  - Sec. 3.1, 3.2, 3.5